

1. Integration and Experiential Learning

Code Otter is a project that I began on my own and later developed with another student, Paul Bauer, as our submission to the Showker Prize in Spring 2024. We designed Code Otter to solve three problems: providing an interactive, self-paced environment for students to learn web development, hosting our bi-weekly JavaScript Challenges run by the Computing Club, and offering a quick, browser-based editor for students to write code and follow along during club talks without needing to install anything. We placed first in the Showker Prize and have continued expanding its features and content ever since.

Code Otter was originally built from scratch and used a simple server and database. After taking several upper-level courses, I revisited the project last summer and refined it using the new tools and practices I had learned. The Server-Side Web Development class taught me how to properly structure a server and manage a MongoDB database which I incorporated some of those techniques into Code Otter to make it more efficient and maintainable. Client-Side Web Development introduced Vue.js, which is a professional framework to make dynamic websites faster and more reliable. Through this process I learned how to manage and develop a full-stack application, designing both the website and the database components and ensuring they work together correctly and consistently. Since students had to make accounts, I had to consider security and proper management of user data, as well as careful feature deployment so that new additions wouldn't break existing features. I also strengthened my skills in modularity to allow Paul and me to work on the app without interfering with each other's work. While my future

career may not be lead developer, learning how to build and maintain an application end-to-end has been invaluable and aligns directly with my major and future goals.

Beyond the technical experience, I also saw Code Otter being used in a real classroom. Early on, Professor Alabanza asked to use it in his Web Design course to help teach introductory web development. At the time, I had written four of the seven lessons currently available. These lessons guide students through concepts with a virtual tutor that demonstrates each step, allowing students to follow along, mimic the tutor, or explore independently without getting lost. The lessons use simple language and small examples to be easy to understand. Professor Alabanza told us that students were asking fewer questions because they understood the material more easily, and he even stopped requiring in-person labs because students could complete them independently. Several students told me personally that Code Otter helped them finally understand the basics and made it easier to move forward.

This semester, a new professor for Web Design adopted Code Otter as a required part of the course. So far, she reported that it's been working smoothly. Currently, Paul and I have been developing new features such as class codes and progress tracking to help with her teaching.

Working on Code Otter has been one of the most meaningful experiences of my college career. It's a large, multifaceted project used in real academic and club settings, and it allowed me to deepen my technical skills while also learning leadership, collaboration, and long-term project maintenance.

Artifact link: <https://claebcode.github.io/portfolio450/1-integration.html>

Last semester, the Computing Club also hosted a “Wack-a-thon,” a wacky-themed hackathon originally intended as a team competition. Not enough people signed up so it became an individual challenge among five participants. I won second place overall and first place in the popular vote. Code Otter was even used in this event, providing a place for everyone to upload their submissions and view everyone else’s after the competition was over. The experience was exhilarating with six hours to design, build, and test a project from scratch. It taught me how to focus under pressure, prioritize tasks, and rely on solid, reliable techniques rather than experimenting with risky ideas. It also reinforced the importance of the “minimum viable product”: a complete, functional project is more valuable than an ambitious one that fails to meet essential requirements. I tend to dream big, and learning to scale back and deliver something polished within constraints was a valuable challenge.

Artifact link: <https://claebcode.github.io/portfolio450/1-integration.html#artifact2>

2. Engaging Diverse Perspectives

All Computer Science majors take a final course called Software Engineering, where students are placed into randomly assigned teams to simulate real-world development environments. Before this class, I had only worked in groups of two at most, and I was used to planning and executing projects largely on my own. Working in a five-person team required me to slow down, listen, and genuinely consider others’ ideas.

Our project was to propose, design, and build a Meal Planning website. During the planning phase, each team member suggested ideas, and we discussed the pros and cons of each option to reduce bias and reach a common decision. While this takes more time and energy than choosing an idea myself, this taught me the value of collaborative decision-making which is an essential skill in the professional world where I won't always be the one in charge.

Our team included students from different cultures and backgrounds which each brought unique perspectives to the project. At first, it was difficult to let go of my own ideas, but over time I learned to appreciate and adapt to different ideas the others brought. Two of us were stronger programmers: I excelled at logic and project structure, while another teammate excelled at design and user experience. Instead of trying to do everything myself, I learned to trust his design instincts while focusing on the areas where I was strongest. We each contributed to all parts of the app, but we divided responsibilities in ways that mirrored real-world development teams.

I'm someone who is naturally shy and often anxious in group settings, and initially I was terrified to start this project. But as the semester went on, I became more comfortable communicating with my teammates, and by the end we were genuinely sad that the project was over. My communication skills improved, and I became more open to feedback and alternative approaches. These experiences will directly support my future career in web development, where teamwork, communication, and adaptability are essential.

Artifact link: <https://claebcode.github.io/portfolio450/2-diverse.html>

3. Public Discourse

For the past three years, Bridgewater has participated in the International Collegiate Programming Contest (ICPC). Each year, we formed teams and practiced weekly using the Kattis problem-solving website. ICPC problems are complex and solving them collaboratively was initially difficult for me because I wasn't the greatest at working through problems with others.

During the competitions each team is given five hours to solve as many problems as possible using only one computer and keyboard. For our first year, our strategy was for each person to independently search through the packet of questions for a solvable problem, solve it on their own, then bring it to me since I was the fastest typist and most experienced programmer of the team. We solved three problems, which wasn't bad, but we knew we could do better.

The next two years we tried a more collaborative strategy. We skimmed the entire packet first to identify the easiest problems, solved them individually, and submitted those quickly. We then ranked the rest and tackled them collaboratively. One person brainstormed ideas, one created test cases, and one typed and tested the code. This strategy doubled our performance and we were able to solve six problems in both later competitions.

Developing a group strategy improved our communication, efficiency, and ability to work through and solve complex problems under pressure. These experiences strengthened my problem-solving skills and helped prepare me for future collaborative work.

Artifact link: <https://claeetcode.github.io/portfolio450/3-public.html>

Beyond ICPC, I also led many Computing Club meetings and events. For three years, I organized and ran our bi-weekly JavaScript Challenges, where we would create small projects ranging from fancy buttons to analog clocks. We created these events to encourage students to code outside of class while also being able to share their creativity in an inspiring way. At each meeting, participants presented their submissions and explained parts of their code. When we first started, I was definitely anxious presenting in front of others, but over time our group bonded as we could see the styles, strengths, and weaknesses of each other. It allowed me to present in front of them with something I was confident in, proud of, and knowledgeable about, compared to a dull presentation from a Gen-Ed class. This greatly helped my presenting skills and helped me become more confident in it.

I also gave three Lightning Talks, which are presentations given by a student or faculty member about a Computer Science or coding topic. Over two years ago, I did my first talk, and there were some complications because not everyone had installed a required program beforehand. Once Code Otter became robust enough, we used it for future talks, eliminating the need to install a program before coming and making the presentations smoother. Besides following along, Code Otter was also used to make submitting and viewing submissions of our JavaScript

challenges easier. Eventually I didn't have any fear standing up in front of the room presenting about these topics. Dr. McGregor and Professor Cullip complemented me on my presentation style and clarity, and Professor Cullip even suggested that I should consider teaching.

Out of everything I did at Bridgewater, these Computing Club presentations helped me the most in combating my fear of public speaking. I discovered that I actually enjoy teaching and presenting, especially when I'm sharing something I'm passionate about. It helped strengthen my skills in explaining complex and technical concepts clearly and comfortably.

Artifact link: <https://claeocode.github.io/portfolio450/3-public.html#artifact2>

4. Global Citizenship and Intercultural Competencies

When I'm not working on coding projects, my main hobby is playing piano and composing music. For my world culture credit, I took a course called Music of Latin America. I entered the class with some prejudices and I thought I wouldn't enjoy Latin American music based on the limited examples I knew. But as we listened to music from different countries, genres, and time periods, I realized that I was wrong and actually really liked some of the music.

One song that really stood out to me was *Soledad y el Mar* by Natalia Lafourcade. I found it so beautiful that I even listened to it regularly outside of class. This experience helped me recognize my own cultural biases and taught me to be more open to other cultures. It also broadened my appreciation for global music and made me more curious about other cultures.

The class even influenced my own compositions. Without consciously trying, I began incorporating subtle Latin American elements into my music such as more syncopated drum patterns which make it feel more alive, melodies leaning towards minor keys which makes it feel more sad, and emotional feel similar to bolero which is the genre of the song listed above. The course wasn't focused on music notation, but more on immersion which naturally influenced some of my creative thoughts.

This class taught me to be more open-minded, curious, and appreciative of cultural differences. These skills are essential for the professional world as the professional world gets more diverse and I work with people from other cultures.

Artifact link: <https://claebcode.github.io/portfolio450/4-global.html>

5. Ethical Reasoning

In my Server Side Web Development course, the final project paired students based on their performance. I earned nearly a perfect score and was paired with a classmate who had been struggling. The project required us to alternate typing on the same computer, ensuring that both partners contributed equally.

This created an ethical challenge: I wanted to do well on the assignment, but I also wanted to support my partner's learning. I knew I could complete the project quickly on my own, but I decided at the start that we were going to work fairly, even if it took longer.

We met multiple times outside of class to ensure we had enough time. My partner started with the easiest task, and I guided him without giving away answers. He would ask me lots of questions and I would explain them carefully so that he would hopefully arrive at the solution himself. Sometimes, despite multiple explanations, he still struggled, and I had to tell him what to do, but I refrained from typing during his turn and tried to talk him through each step. During my turns, I was tempted to quickly type everything out, but I knew that wouldn't help him understand. So, I typed slowly and explained my reasoning so he would follow along. After each section, I would ask him what potential issues might arise in our code, and then we tested it together.

Overall, this was a challenging experience balancing fairness, patience, and responsibility. It taught me how to mentor someone respectfully while still meeting project requirements. Going forward, I hope to approach future collaborations with the same amount of commitment, fairness, and support.

Artifact link: <https://claebcode.github.io/portfolio450/5-ethical.html>